



# RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Civil Engineering  
Academic Year 2022–2023 (Odd Semester)

► Degree, Semester & Branch: III semester B.E Civil Engineering

Course Code & Title: CE3301 & Fluid Mechanics

Name of the Faculty member (s): Mrs.R.Kalaimani

## Innovative Practice Description

- **Unit / Topic:** Determination of friction factor in pipes
- **Course Outcome:** CO4
- **Topic Learning Outcome:**
  - ✓ Determine whether a pipe flow is laminar or turbulent flow for specified flow conditions
  - ✓ Calculate the Reynolds number for pipe flow with specified flow conditions
  - ✓ Calculate the entrance length for pipe flow with specified flow conditions
- **Activity Chosen:** Virtual Laboratory
- **Justification:**
  - ✓ Apply the basic engineering concept in friction factors in pipes
  - ✓ Explain the working of friction factors in pipes
- **Time Allotted for the Activity:** 15 minutes
- **Details of the Implementation:**  
44 students are participated totally in online virtual simulation lab. Virtual lab also enabled to
- **CO – PO / PSO mapping:**

| CO  | PO1 | PO2 | PO9 | PSO2 |
|-----|-----|-----|-----|------|
| CO4 | 3   | 1   | 1   | 3    |

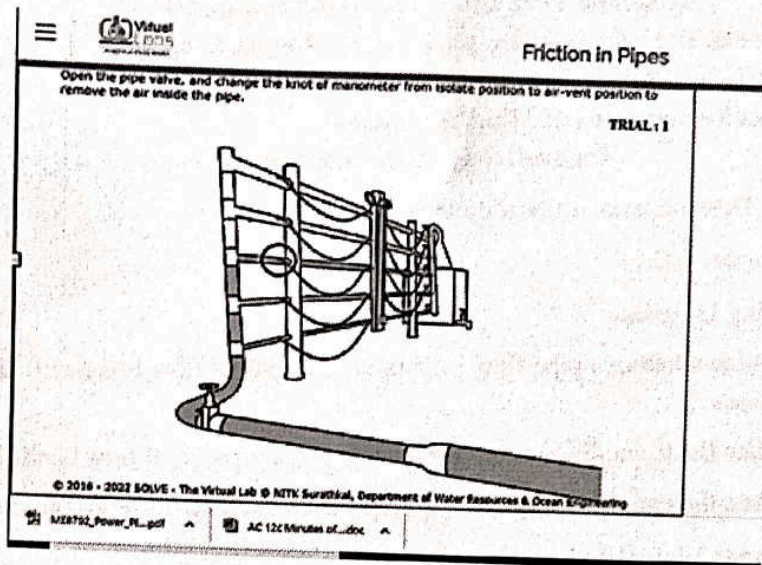
(1 – Low      2 – Moderate      3 – High)

- **PO / PSO mapped:**

| Innovative practice                  | PO1                                                                                                                                 | PO2                                                                                     | PO9                                                                               | PSO2                                                             |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------|
|                                      | 3                                                                                                                                   | 1                                                                                       | 1                                                                                 | 3                                                                |
| <b>Justification for correlation</b> | The mechanics behind pipe flow and the energy losses associated with it provides a fundamental engineering knowledge to the student | Analysis of flow through pipes is a promising problem solving area in Civil Engineering | Demonstrate the ability to define a complex problem in frictional losses in pipes | Calculate the major and minor losses at different flow condition |



- Images / Screenshot of the practice:



- Reflective Critique:

- ❖ *Feedback of practice from students and other stakeholders:*

The overall feedback from the students were really good and students felt that it is very easy to understand the real time operation of friction factor in pipes

- ❖ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

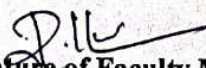
- After completing this hand on session in virtual laboratory, the students were able to understand the friction factor in pipes very clearly.
- This virtual simulation also helps the students to know the stability of the system.

- ❖ *Challenges faced in implementation:*

The output of simulation part can be better.

References:

- ❖ <http://fm-nitk.vlabs.ac.in/List%20of%20experiments.html>

  
Signature of Faculty Member

  
HOD